

DESIG Syllabus

David Shamlin, June 2026

This course introduces you to digital technology concepts related to typical consumer usage scenarios. The primary aim of this course is to expand your understanding of foundational digital technology (i.e, hardware, software, the Internet, and the Cloud) and introduce you to related skills to better enable you in your ongoing learning.

Learning Outcomes

Upon completing this course, you will be able to:

1. Identify the basic components found inside digital devices (like computers/laptops, smartphones, and tables) and describe their purpose.
2. Describe what software is and how it works in general terms as identify kinds of software you use.
3. Have familiarity with common digital “tools” (i.e., particular types of apps like browsers, email clients, and search engines) and their key features.
4. Assess your digital security & privacy needs and pursue strategies for increasing your digital security & privacy based on your personal needs/preferences.

Outline

Note: key topics that I think merit more time/attention during our “second round” of the course are (1) data, (2) software, and (3) user interface. As we progressed through our SIG sessions in 2026, there were a significant number of moments where I thought “this would be easier for me to explain and you to understand if I had explained data and software in more detail during the Fall 2025 course”. Regarding the user interface topic, our previous discussions may have been sufficient but I am not convinced of that yet; I strongly believe having a strong grasp on UI-related concepts/constructs is central to being able to effectively use software. While I don’t think it’s too hard to develop a “surface-level” understanding of UI concepts, we may benefit from deeper exploration in a second round—and possibly tying it to the additional software concepts/constructs

1. Foundation technology

1. Hardware

1. Processor, memory, & storage
2. Connectivity: ports/cables, WiFi, cellular, and bluetooth
3. Interactivity: screen, speakers, keyboard, mouse/trackpad/touchscreen, speakers, mic, and camera
 1. Gestures: the basics of using a mouse/trackpad/touchscreen
4. Data: what is *binary data* and how is the size of it measured (i.e., KB, MB, GB, and TB)?

5. Sizing: how the processing power of digital devices is measured
2. Software
 1. What is software?
 2. **NEW:** Software types (e.g., apps, web apps, extensions, daemons, etc)
 3. **NEW:** App states (i.e., software process life-cycle)
 4. **NEW:** What happens when you install/update software?
3. Operating Systems
4. User Interface (UI)
5. File systems
6. Settings
7. Data: What it is
 1. **NEW:** Data types: numerics, text, date/time
 2. **NEW:** Data structures: records, lists, dictionary, trees, tables, graphs
 3. Files
2. Core technology
 1. The Internet
 2. The Web
 3. The Cloud
 4. **NEW:** Data: How it behaves
3. Privacy & security
 1. Bad actors: hackers
 2. Common threats:
 1. Bad actors (i.e., hackers)
 2. Scams, phishing, and malware
4. Common tools
 1. Email
 2. Browser
 3. **NEW if there is interest:** Word processors
 4. **NEW if there is interest:** Spreadsheet
 5. **NEW if there is interest:** Presentation software
 6. **NEW if there is interest:** Video conferencing software

7. **NEW:** *Are there are apps you commonly use or you would like to use that we should consider adding?*
8. Search engines & (**NEW**) recommendation engines
9. Password managers
10. Virtual Private Networks (VPNs)
5. Controversies: where and how digital technologies impact society and culture for better or worse
 1. Cyber-security / Cyber-warfare: how digital technology is affecting geopolitical stability
 2. **NEW:** Misinformation, disinformation, deepfakes
 3. **NEW:** Social media
 1. Mental health concerns: impacts on attention span, overall cognition, independent problem solving, and patience
 2. Content moderation: freedom of speech/expression vs legal/ethical challenges (e.g., misinformation, hate speech, privacy violations)
 3. Social media personality types: influencers & trolls
 4. Cyber-bullying
 4. **NEW:** Artificial intelligence

On a couple occasions in our sessions, folks have made emphatic statements of not wanting AI on their devices (which I empathize with.) The fact of the matter is you are using AI at times. A lot of the societal/political concerns being broadly discussed today relate to a relatively new type of AI. My intent in suggesting a lesson on AI is to describe what “artificial intelligence” is in practical terms and outline the types/classes/types of AI in order for you to better understand its pros & cons (most specifically risks related to its use and possible impacts on humanity, society, the environment, etc.)

 1. What is “artificial intelligence”
 2. Types of AI
 3. Application of AI (i.e., where do you encounter AI when using your device(s) and/or using online services)
5. **NEW:** Cryptocurrencies

While I don’t think anyone in the group is currently using or interesting in acquiring cryptocurrency, I am open to including a session on it if there is interest. Cryptocurrency

has socio-economic, political, and environmental impacts—not all of them good in my opinion. It could be interesting/valuable to have a session where we explore this digital technology just enough to enhance our abilities to be good “digital citizens”. Also, the central technology underlying cryptocurrency (i.e., blockchain) has some capabilities with potential to revolutionize digital privacy.